# Assessment Report on *"Matrix Theory Reloaded: A BPS Road to Holography"*

**Article under review:** C. D. A. Blair, J. Lahnsteiner, N. A. Obers, and Z. Yan, *"Matrix Theory Reloaded: A BPS Road to Holography,"* IFT-UAM/CSIC-24-139, NORDITA 2024-033 (`paper.tex`).

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## 1 Principal Achievement of the Paper

The principal achievement of this paper is the unification of three foundational concepts in string and M-theory—the BPS decoupling limits defining matrix gauge theories on D$p$-branes, the bulk near-horizon decoupling limits underlying the AdS/CFT correspondence, and $T\bar{T}$-like stress-tensor deformations—within a single target-space geometric and U-duality framework. By treating the matrix-theory decoupling limit not merely as a worldvolume limit of D$p$-branes in flat space, but as a self-consistent BPS limit of the full ten-dimensional type II superstring background (termed *Matrix p-Brane Theory*, or M$p$T), the authors resolve a longstanding conceptual tension between matrix theory and holography and establish three major results:

(a) **Geometric unification of matrix-theory and near-horizon limits (Resolving Question 2):** In the M$p$T prescription (Section 2), keeping $\alpha'$ fixed while introducing a dimensionless control parameter $\omega \to \infty$, the ten-dimensional Lorentzian metric $G_{\mu\nu}$, dilaton $\Phi$, and Ramond-Ramond (RR) $(p+1)$-form $C^{(p+1)}$ are parameterized as

$$G_{\mu\nu} = \omega\, \tau_{\mu\nu} + \omega^{-1} E_{\mu\nu}\,, \qquad e^{\Phi} = \omega^{\frac{p-3}{2}} e^{\varphi}\,, \qquad C^{(p+1)} = \omega^2 e^{-\varphi} \tau^0 \wedge \cdots \wedge \tau^p + c^{(p+1)}\,, \tag{1}$$

where the critical RR potential cancels the D$p$-brane tension and the target space degenerates into a non-Lorentzian codimension-$(p+1)$ foliated $p$-brane Newton-Cartan geometry $(\tau_\mu{}^A, E_\mu{}^{A'})$ governed by $p$-brane Galilean boosts and an emergent local dilatation symmetry ($\tau^A \to \Delta^{1/2}\tau^A$, $E^{A'} \to \Delta^{-1/2} E^{A'}$, $e^{\varphi} \to \Delta^{(p-3)/2} e^{\varphi}$). The authors' key insight (Section 3.1) is that applying this *exact same* M$p$T prescription at the asymptotic infinity ($R \to \infty$, $H \to 1$) of a curved, backreacted D$p$-brane supergravity solution rescales $R = \omega^{-1/2} r$ and $L^{7-p} = \omega^{(p-3)/2} \ell^{7-p}$, transforming the harmonic function into

$$H(R) = 1 + \left(\frac{L}{R}\right)^{7-p} = 1 + \omega^2 \left(\frac{\ell}{r}\right)^{7-p}. \tag{2}$$

Sending $\omega \to \infty$ automatically drops the "1" in $H$, inducing the bulk Maldacena/IMSY near-horizon limit. Crucially, this explains why $(p+1)$-dimensional maximally supersymmetric Yang-Mills (SYM) couples to a non-Lorentzian M$p$T geometry in the weak-coupling open-string regime yet is dual to a *Lorentzian* near-horizon geometry (such as $\mathrm{AdS}_5 \times S^5$ for $p = 3$) at strong coupling: the near-horizon bulk metric

$$\mathrm{d}s^2 = \mathbb{\Omega}(r)\, \eta_{AB}\, \mathrm{d}x^A \mathrm{d}x^B + \mathbb{\Omega}(r)^{-1} \left(\mathrm{d}r^2 + r^2 \mathrm{d}\Omega^2_{8-p}\right), \qquad \mathbb{\Omega}(r) = \left(\frac{r}{\ell}\right)^{\frac{7-p}{2}}, \tag{3}$$

geometrically realizes the non-Lorentzian M$p$T limit as $r/\ell \to \infty$, with the radial warp factor $\mathbb{\Omega}(r)$ playing the role of the decoupling parameter $\omega$ up to the emergent local dilatation symmetry.

(b) **Systematic generation of non-Lorentzian holography and the DLCQ$^n$/DLCQ$^m$ classification (Addressing Question 1):** By uplifting M0T to the Discrete Light Cone Quantisation (DLCQ) of M-theory and mapping the null circle via U-duality to a spatial circle in *non-relativistic M-theory* (the critical M2-brane limit with membrane Newton-Cartan geometry, and its magnetic M5-brane dual), the authors organize all BPS decoupling limits into U-duality orbits of iterated DLCQs (Section 4) and derive the maximally extended 32-supercharge M0T superalgebra via İnönü-Wigner contraction (Section 2.3). They conjecture that top-down holography in string/M-theory is captured by a DLCQ$^n$/DLCQ$^m$ ($m > n$) correspondence, where $m - n$ asymptotic BPS limits eliminate the constant "1" in $m - n$ bulk harmonic functions without altering the Lorentzian/non-Lorentzian character of the bulk geometry:

- $n = 0, m = 1$ (DLCQ$^0$/DLCQ$^1$) encompasses standard AdS/CFT (D3, M2, M5) and IMSY D$p$-brane dualities;
- $n = 0, m = 2$ (DLCQ$^0$/DLCQ$^2$) captures intersecting-brane dualities such as the D1-D5 $\mathrm{AdS}_3/\mathrm{CFT}_2$ system via a multicritical M1T–M5T limit;
- $n = 1, m = 2$ (DLCQ$^1$/DLCQ$^2$) systematically generates novel non-Lorentzian holographic dualities (Sections 3.2–3.3, Table 1) by applying an off-aligned asymptotic M$p$T limit to a D$q$-brane geometry satisfying the $\frac{1}{4}$-BPS finite-mass condition $m + n = 4$, followed by a near-horizon M$q$T limit, thereby unifying and generalizing recent bottom-up/top-down proposals in the literature.

(c) **Intrinsic bulk reconstruction and higher-dimensional $p$-brane $T\bar{T}$ flow equations (Addressing Question 3):** Viewing the BPS decoupling limit in reverse ($\boldsymbol{t} = \omega^{-2} > 0$), the authors show through S- and T-duality from type IIB non-relativistic string theory (Section 5) that reintroducing $\omega^{-2}$ corresponds to a $p$-brane generalization of the 2D $T\bar{T}$ deformation. Promoting $\boldsymbol{t}$ to the coordinate-dependent profile $\boldsymbol{t}(r) = [\mathfrak{Q}(r)]^{-2}$ reconstructs the curved near-horizon bulk geometry (3) directly as a deformation of the flat asymptotic M$p$T geometry. At the worldvolume level (Section 6), this yields a closed-form stress-tensor flow equation in arbitrary dimension $d = p + 1$ that deforms a free multi-scalar theory into the gauge-fixed Dirac-Nambu-Goto action,

$$\frac{\partial \mathcal{L}(\boldsymbol{t})}{\partial \boldsymbol{t}} = \frac{1}{2\boldsymbol{t}^2}\left\{\mathrm{tr}(\mathbb{1} - \boldsymbol{t}\mathcal{T}) - (p-1)\left[\det(\mathbb{1} - \boldsymbol{t}\mathcal{T})\right]^{\frac{1}{p-1}} - 2\right\}, \tag{4}$$

as well as exact closed-form flow equations for abelian gauge-plus-scalar theories to the D$p$-brane DBI action for $p = 1$ and $p = 2$ (Eq. 6.53). Notably, for $p = 1$ the flow explicitly requires $\sqrt{-\det \mathcal{F}}$, explaining why the standard 2D $T\bar{T}$ deformation of Maxwell theory fails to produce the D1-brane DBI action, while for $p = 2$ the D2-brane DBI flow coincides identically with the $p = 2$ membrane flow $\frac{\partial \mathcal{L}}{\partial \boldsymbol{t}} = \frac{1}{4}[\mathrm{tr}(\mathcal{T}^2) - (\mathrm{tr}\,\mathcal{T})^2 + 2\boldsymbol{t}\det\mathcal{T}]$ by virtue of 3D scalar–vector Hodge duality.

## 2 Principal Drawback of the Paper

The principal drawback of the paper is that **its extensions beyond standard single-stack relativistic holography—both to non-Lorentzian DLCQ$^1$/DLCQ$^2$ bulk geometries and to intrinsic bulk reconstruction via $p$-brane $T\bar{T}$ deformations—rely on kinematical scaling prescriptions and ad hoc workarounds rather than dynamically verified bulk supergravity or quantum/non-abelian field-theoretic control.** This overarching limitation manifests in three specific technical and physical weaknesses:

(a) **Ad hoc manipulations (smearing vs. extrinsic $N$-rescaling) and unverified supergravity equations in DLCQ$^1$/DLCQ$^2$ bulk geometries (Section 3.3):** When

an off-aligned asymptotic M$p$T limit is applied to a bulk D$q$-brane solution with $n > 0$ transverse directions of the D$q$-brane lying along the M$p$T longitudinal sector (subject to $m + n = 4$), the harmonic function reads

$$H = 1 + \omega^{\frac{p+q-10}{2}} \left( \frac{\ell}{\sqrt{x^u x^u + \omega^{-2} x^i x^i}} \right)^{7-q} . \tag{5}$$

Except in the special cases $n = 0$ ($q = p+4$, such as D4 in M0T) or $p+q = 10$ (such as D3 in M7T), the intrinsic M$p$T prescription *fails* to produce a finite, nontrivial harmonic function as $\omega \to \infty$. Consequently, for almost the entire catalogue of $p, q \leq 4$ configurations in Table 1, the authors must modify the limit by hand using one of two mutually incompatible procedures:

(i) *Smearing* the D$q$-brane across the $n$ M$p$T-longitudinal directions $x^u$, which yields $H = 1 + (\boldsymbol{\ell}/\|x^i\|)^{7-q-n}$ localized solely in the M$p$T-transverse coordinates $x^i$; or

(ii) Introducing an *extrinsic rescaling of the brane number* $N \to \omega^{3-q+m} N$ (i.e. $\ell^{7-q} \to \omega^{(10-p-q)/2} \ell^{7-q}$), which yields $H = 1 + (\ell/\|x^u\|)^{7-q}$ localized solely in the M$p$T-longitudinal coordinates $x^u$ (and for $p = q$ requires the unphysical scaling $N \to \omega^{-2} N \to 0$).

As the authors acknowledge in Sections 3.3.2 and 7, the $N$-rescaling lies outside the intrinsic definition of the M$p$T limit and lacks a clear justification in terms of physical brane sources. Worse still, this ambiguity leaves two distinct candidate bulk geometries for each entry in Table 1 without determining which (if either) is the true holographic dual. Most critically, **the authors do not formulate the ten-dimensional M$p$T supergravity equations of motion or torsional constraints, nor do they verify whether either family of limiting backgrounds actually solves them or preserves supersymmetry.**

(b) **Kinematical, classical, and abelian-only nature of the $p$-brane $T\bar{T}$ flow and bulk reconstruction (Sections 5–6):** Although Section 5 frames the generation of the bulk near-horizon geometry (3) as an "intrinsic" $p$-brane $T\bar{T}$ deformation of the flat asymptotic M$p$T geometry, this reconstruction is purely kinematical: the radial profile $\mathbb{\Omega}(r) = (r/\ell)^{(7-p)/2}$ is not derived from solving a renormalization-group or stress-tensor flow equation, but is imported from the known supergravity harmonic function. Furthermore, promoting a constant worldvolume deformation parameter $\boldsymbol{t} = \omega^{-2}$ to a target-space function $\mathbb{\Omega}(X)^{-2}$ is only justified as a marginal worldsheet operator $(-\frac{1}{4\pi\alpha'} \int \mathrm{d}^2\sigma \sqrt{-h}\, \mathbb{\Omega}(X)^{-2} \lambda\bar{\lambda})$ in the Polyakov formulation of the $p = 1$ fundamental string (and for the $p = 0$ worldline without gauge fields). For $p > 1$, no Polyakov-like marginal formulation is constructed, and in $d = p+1 > 2$ dimensions the composite operators in Eq. (4) are non-renormalizable and lack Zamolodchikov's 2D quantum factorization property. On the worldvolume side (Section 6), the derived flow equations are restricted to the **abelian ($U(1)$) bosonic sector**: they do not capture the non-abelian commutator potential $[X^i, X^j]^2$ or fermionic terms of matrix gauge theory (Eq. 2.66), which are essential for large-$N$ holography. Even within the abelian bosonic sector with $F_{\alpha\beta} \neq 0$, closed-form flow equations are found only for $p = 1, 2$; for the central holographic case $p = 3$ ($\mathrm{AdS}_5/\mathrm{CFT}_4$), the flow equation can only be obtained perturbatively to $O(\boldsymbol{t}^1)$ (Eq. 6.25).

(c) **Breakdown of multi-direction T-duality with background fluxes (Section 2.5.2):** In deriving the curved-space non-abelian matrix gauge theory action from Myers' dielectric brane action (Eq. 2.66), finiteness of the M$p$T limit and compatibility with T-duality from D9-branes require imposing by hand the restrictive conditions $B_{ij} = 0$ and $\tau_{ij} = \tau_{i\alpha} = 0$ (Eq. 2.63). Because transverse $B$-fields play a central role in non-commutative Yang-Mills (NCYM) limits (as seen for the D2-brane in M0T in Section 2.5.1), this restriction prevents the paper's T-duality Buscher rules and curved matrix actions from capturing general flux backgrounds.

# 3 Most Worthwhile Research Idea Building on the Paper

## Evaluation of Candidate Directions

The paper opens three natural avenues for follow-up research:

1. **Dynamical Supergravity and Supersymmetry Verification of DLCQ$^1$/DLCQ$^2$ Non-Lorentzian Holographic Backgrounds:** Testing the smeared vs. large-$N$ rescaled brane geometries of Section 3.3 against the field equations, intrinsic torsional constraints, and Killing spinor equations of non-Lorentzian supergravity (via the 11D M-theory uplift of Section 4.2 and 10D reductions), and extending the M$p$T duality rules to $B_{ij} \neq 0$.

2. **Timelike T-Duality to Carrollian Brane Limits ($p \leq -1$) and de Sitter / Flat Holography:** Applying the asymptotic BPS decoupling framework to Hull's type II$^*$ string theory, linking M($-1$)T (IKKT matrix theory on D-instantons) and Euclidean E$p$-branes to Carroll-like target space geometries and $\mathrm{dS}_5/\mathrm{CFT}_4$ near-horizon limits (suggested in Footnote 6 and Section 7).

3. **Exact $d = 4$ ($p = 3$) DBI and Non-Abelian $p$-Brane $T\bar{T}$ Flows via Auxiliary-Field Formulations:** Using Howe-Tucker/Polyakov auxiliary metrics or Courant-Hilbert/Cayley-Hamilton invariants to close the $p = 3$ DBI flow equation non-perturbatively and extend it to the non-abelian/supersymmetric sector.

While Direction 2 offers an exciting conceptual bridge to de Sitter and Carrollian holography and Direction 3 is valuable for classical nonlinear electrodynamics, **Direction 1 best balances high feasibility with transformative impact**. It directly resolves the paper's most acute physical deficiency—the unverified status and smearing vs. $N$-rescaling ambiguity of the DLCQ$^1$/DLCQ$^2$ non-Lorentzian bulk geometries—using supergravity tools that are already fully developed and available.

## Proposed Research Project: Supergravity Consistency, Supersymmetry, and Torsional Constraints of Non-Lorentzian Holographic Brane Geometries

**Core Objective.** Establish which of the candidate DLCQ$^1$/DLCQ$^2$ bulk geometries in Section 3.3 (smeared vs. large-$N$ rescaled) are genuine, supersymmetric solutions of non-Lorentzian supergravity, determine the precise superconformal/Killing spinor algebras they preserve in the near-horizon limit, and generalize the M$p$T T- and S-duality transformations to non-vanishing transverse $B$-fields ($B_{ij} \neq 0$).

**Concrete Methodology and Feasibility.** This project is highly feasible because it does not require constructing ten-dimensional M$p$T supergravity for every $p$ from scratch. Instead, one can exploit two key structural results already established in `paper.tex` and the recent literature:

(i) **Uplift to 11D Non-Relativistic Supergravity:** As shown in Section 4.2, every 10D M$p$T limit ($p \leq 5$) is related by dimensional reduction and T/S-duality to the M2-brane BPS decoupling limit defining 11D *non-relativistic M-theory* (Eq. 4.12, with membrane Newton-Cartan geometry $\tau_\mu{}^A$ ($A = 0, 1, 2$) and $E_\mu{}^{A'}$ ($A' = 3, \ldots, 10$)) or its magnetic M5-brane dual (Eq. 4.15). Crucially, the complete bosonic action, equations of motion, Killing spinor equations, and intrinsic torsional constraints of 11D non-relativistic supergravity have recently been constructed in Ref. [94] (Bergshoeff, Blair, Lahnsteiner, and Rosseel, `arXiv:2407.21648`). In particular, consistency of 11D non-relativistic supergravity requires torsional constraints on $\tau_\mu{}^A$, which admit two distinct consistent branches: a *maximally supersymmetric* branch (preserving all 32 supercharges in the vacuum) and a *half-maximally supersymmetric* branch (preserving 16 supercharges).

(ii) **Testing Smeared vs. $N$-Rescaled Brane Solutions:** Under the M-theory uplift dictionary of Section 4.2, the $p, q \leq 4$ D$q$-brane configurations in M$p$T (Table 1) uplift to off-aligned M2-brane and M5-brane supergravity solutions subjected to the asymptotic M2-brane decoupling limit (specifically: a bulk M2-brane overlapping the M2-limit along $1+0$ directions with $m=2, n=2$, and a bulk M5-brane overlapping the M2-limit along $1+1$ directions with $m=4, n=1$). One can therefore directly substitute both the *$N$-rescaled* (longitudinally localized, $H = 1 + (\ell/\|x^u\|)^k$) and *smeared* (transversely localized, $H = 1 + (\boldsymbol{\ell}/\|x^i\|)^{k'}$) ansätze into the 11D non-relativistic supergravity equations of motion, torsional constraints, and Killing spinor equations:

- Because the longitudinal vielbein takes the form $\tau^a = H^{-1/6}\mathrm{d}x^a$, $\tau^u = H^{1/3}\mathrm{d}x^u$, dependence of $H$ on the transverse coordinates $x^i$ (the smeared branch) vs. the longitudinal coordinates $x^u$ (the $N$-rescaled branch) produces qualitatively different intrinsic torsion components $\mathrm{d}\tau^A$.
- Solving the Killing spinor equations before and after the second ($\tilde{\omega} \to \infty$) near-horizon limit will unambiguously discriminate between the two branches: it will determine whether the $N$-rescaled branch satisfies the maximally supersymmetric torsional constraints (and exhibits standard $\frac{1}{2}$-BPS $\to$ maximal superconformal enhancement in the near-horizon limit), while the smeared branch is restricted to the half-maximal torsional branch (or vice versa).

(iii) **Incorporating Transverse Fluxes ($B_{ij} \neq 0$) and Duality Asymmetry:** Finally, by relaxing the gauge-fixing assumption $B_{ij} = 0$ (Eq. 2.63) in the finite-$\omega$ Buscher rules of Section 2.4 prior to taking $\omega \to \infty$, one can systematically classify how T- and S-duality act on M$p$T backgrounds with non-trivial $B$-fields and RR fluxes. When $B_{ij} \neq 0$ along a transverse two-torus, the inverse metric block $(G_{ij} + B_{ij})^{-1}$ alters the $\omega$-scaling of the dual metric so that T-duality on two transverse directions maps M$p$T to M$p$T (preserving the foliation codimension) rather than M$(p+2)$T. Tracking this "duality asymmetry" directly connects the M$p$T framework to Morita equivalence in NCYM (Section 2.5.1) and generates new flux-supported DLCQ$^1$/DLCQ$^2$ holographic backgrounds without requiring ad hoc $N$-rescalings.

**Expected Impact.** Executing this project will place the paper's DLCQ$^n$/DLCQ$^m$ holographic conjecture on a rigorous dynamical foundation. First, it will resolve the physical puzzle of Section 3.3.2 by proving from first principles (via supergravity torsional constraints and supersymmetry preservation) why the longitudinally localized ($N$-rescaled) and transversely localized (smeared) non-Lorentzian near-horizon geometries belong to distinct supersymmetric sectors of non-Lorentzian supergravity. Second, by explicitly solving the Killing spinor equations in the near-horizon limit, it will provide the exact global supergroups needed to match observables against the dual non-relativistic/moduli-space quantum field theories on the boundary. Third, extending the duality web to $B_{ij} \neq 0$ will seamlessly unify non-commutative gauge/gravity duality with non-Lorentzian holography, completing the foundational program initiated in `paper.tex`.